

A thematic ambiguity account of non-culminating construals of Mandarin Chinese experiencer-object psych verbs

Abstract: We examine in this paper a class of verbs traditionally labelled experiencer-object psych verbs (EO psych verbs) with respect to their culminating entailments in Mandarin Chinese (Chinese); that is, whether the change in mental state these verbs describe come to hold in their experiencer objects. The basic observation is that these predicates optionally culminate when they have subjects that are agentive, whereas non-agentive subjects enforce culmination, in line with what has been observed for (non-psych) accomplishment predicates both within and in many other languages. We propose that non-culmination arises as a result of a thematic ambiguity, where the agentive variants of these verbs have a canonical transitive structure with the surface object interpreted as a theme/patient and not an experiencer, whereas the non-agentive variants have a deep unaccusative structure where the surface object is interpreted as a true experiencer. The lexical entailments of these respective thematic roles, formalized appropriately, will naturally capture the empirical observations regarding culmination. This is contrary to what has been claimed as the source of non-culmination with accomplishments in the language, namely the weak perfective marker *-le* or a sub-lexical modal operator attaching to genuine accomplishment predicates with both agent and non-agent subjects. We show using well-established aspectual diagnostics that EO psych verbs fall into at least two classes in terms of how their agentive uses differ from their non-agentive uses and crucially, that they do not behave alike in terms of aspectual categories both within and across the two classes. Regardless of which class an EO psych verb falls into, there also remains a genuine asymmetry with regard to the surface objects' ability to serve as the antecedent of the reflexive anaphor *ziji*. These therefore serve as independent evidence outside of non-culminating entailments that support an analysis where the agentive and non-agentive variants receive different treatments independent of the classes, which we analyze as a difference in thematic interpretation of the surface object. Previous proposals about non-culminating accomplishment predicates in the language, in virtue of assuming that the agentive and non-agentive variants belong to the same aspectual class, therefore cannot be a general proposal about non-culmination and the properties of individual verb classes need to be taken into account when locating the source of non-culmination.

Keywords: (non-)culmination; experiencer-object psych verbs; Mandarin Chinese; experiencers; patients/themes

1. Introduction

It has been observed that cross-linguistically, non-agentive subjects enforce culmination of an accomplishment predicate (in the sense of Vendler 1967) where the surface object must undergo the change described by the predicate. With agentive subjects, the object can but need not undergo change, i.e., non-culmination or zero-change (e.g., Demirdache & Martin 2015; Martin 2020; a.m.o.). This effect regarding culminating entailments has been observed in accomplishment predicates of a diverse range of languages such as Thai (Koenig and Muansuwan 2000), Korean (Beavers & Lee 2020; Fritz-Huechante et al. 2020), Hindi (Singh 1991,

1998; Altshuler 2013), Mandarin (Demirdache et al. 2017; Liu 2018; Martin et al. 2020, 2021), Indonesian (Sato 2019), as well as many Romance and Germanic languages such as English, albeit for a much more restricted set of verbs (Oehrle 1976; Koenig & Davis 2001; Martin 2020; *a.m.o*). Example (1) below illustrates such an effect in Mandarin Chinese (henceforth Chinese) with the accomplishment predicate *shāo* “burn”.

- (1) a. 张三烧了书，但书没烧着。

Zhāngsān shāo-le shū, dàn shū méi shāo-zháo.

Zhangsan burn-PFV book but book NEG burn-RES.

Intended: ‘Zhangsan burnt the book, but it didn’t get burnt.’

- b. 那场火烧了书，#但书没烧着。

Nà chǎng huǒ shāo-le shū, #dàn shū méi shāo-zháo.

that CL fire burn-PFV book but book NEG burn-RES

Intended: ‘That fire burnt the book, but it didn’t get burnt.’

In addition to regular accomplishment predicates, the effect has also been observed in verbs denoting changes of mental states where the surface object experiences the mental state change (EO psych verbs). The following example illustrates the effect with the French verb *provoquée* “provoke”. Naturally, the logical approach would be to assimilate these cases with regular accomplishment predicates, with the same mechanisms accounting for the distribution of non-culminating entailments (Martin & Schäfer 2017).

- (2) a. *Pierre l’a provoquée, mais cela ne l’a pas touchée du tout.*
 Pierre her has provoked but this NEG her has NEG touched at all

Intended: ‘Pierre provoked her, but this didn’t touch her at all.’

- b. *Cette remarque l’a provoquée, #mais cela ne l’a pas touchée du tout.*
 this remark her has provoked but this NEG her has NEG touched at all

Intended: ‘This remark provoked her, but this didn’t touch her at all.’

(Martin & Schäfer 2017: 89)

Building on this line of work, we examine EO psych verbs in Chinese, an understudied class of verbs in the language with respect to non-culminating entailments. We aim to demonstrate that the contrast between agents and non-agents regarding culmination entailments *cannot* be assimilated to previously proposed general mechanisms accounting for non-culmination of accomplishment predicates since many Chinese EO psych verbs display varied *aktionsart* profiles. Instead, we argue that the non-agentive variants of such verbs are underlyingly unaccusative, where both surface arguments are base-generated low within the verb phrase and that the surface object is assigned a true *experiencer* role. The agentive variant, on the other hand, has a run-of-the-mill transitive structure where the surface object is assigned a canonical

theme/patient role. Non-culminating readings therefore arise as a result of the specific entailments of the different thematic roles assigned to the surface object, formalized in the style of Reinhart (2002) where the thematic role determines whether a change in mental state of its referent is entailed. An experiencer role necessarily entails a change in mental state whereas a theme/patient can but need not entail such a change, leading to the observed distribution of non-culminating readings. We show further that such an analysis is backed up by binding facts regarding the Chinese anaphor *ziji*, famously shown to be sensitive to various factors about the referential properties of its possible antecedents.

We proceed as follows. Section 2 briefly reviews previous approaches to the source of non-culmination in accomplishment predicates and EO psych verbs that assume the agentive and non-agentive variants share the same underlying event structure, an assumption we will refute based on the evidence we will show in section 3. Based on this evidence, we provide the main tenets of our analysis for the distribution of non-culminating readings in EO psych verbs as arising from a thematic ambiguity between the agentive and non-agentive variants, where the surface object receives different thematic roles, mapping either to a transitive or an unaccusative structure. The thematic and structural differences between the agentive versus non-agentive variants of this class of verbs, an analysis already discovered by work on both Chinese (Verhoeven 2010; Cheung & Larson 2015) and cross-linguistically (Landau 2010) provides a straightforward explanation for the distribution of non-culminating readings. Section 4 provides further independent evidence of key differences between the surface objects of the agentive and non-agentive variants in terms of binding of the reflexive anaphor *ziji*: the surface object of the non-agentive variant can bind into the subject containing an anaphor whereas the object of the agentive variant cannot, and the surface object of the non-agentive variant is also more privileged to serve as a binder in a forward binding scenario than that of the agentive variant. These facts therefore indicate a difference in the surface object's prominence on a thematic hierarchy between the agentive and non-agentive variant, a factor that the binding of *ziji* has been shown to be sensitive to. Section 5 points out some implications of the results here and concludes.

2. A general picture of experiencer-object psych verbs and non-culmination

We begin by providing some background on EO psych verbs more generally. It has been noted that EO verbs can be realized in three distinct patterns alternating between agentive, eventive¹, and stative (Grimshaw 1990; Landau 2010; Temme 2018). (3) illustrates an example in English, with an agentive subject in (3a), a non-agentive but animate subject in (3b), and a clausal stative subject in (3c).

- | | |
|--|----------------|
| (3) a. Nina frightened Laura deliberately. | [agentive] |
| b. The noise frightened Laura. | [eventive] |
| c. That she could be president frightened Laura. | [mental state] |

¹ The term 'eventive' in Temme (2018) refers to non-agentive change of state. It is not to be confused with the term 'event' as in 'event structure' or 'event type'. The latter refers to the four Vendlerian types (1967), including activity, accomplishment, achievement, state, characterising the internal temporal structure of an event.

According to Temme (2018), the agentive EO structure (3a) involves only a volitional agent without a change of state, the eventive EO structure (3b) only encodes a change of state, and the mental state reading (3c) must be stative. Landau (2010) argues that while the agentive variants of EO verbs are accomplishments and should not be treated as psych verbs, the non-agentive variants are stative and involve no change of state. As is already implicit in these works, the agentive variants differ from the non-agentive cases in terms of their event structure, though how they differ remains controversial. We will return to such event structural differences in the next section.

Chinese EO psych verbs also fall into two groups depending on whether the experiencer is realized as the subject or the object. In many cases, the EO psych verb does not have a morphologically related counterpart ES psych verb; rather, they are often realized as lexically different verbs even if the real-world concepts they denote are related, as shown in (4).

- (4) a. Zhāngsān de huà_{TH} xià-le Lǐsì_{EXP} yítiào. [EO psych verb]
 Zhangsan MOD word frighten-PERF Lisi a.jump
 'Zhangsan's words frightened Lisi.'
- b. Lǐsì_{EXP} pà chóngrǐ_{TH}. [ES psych verb]
 Lisi fear bug
 'Lisi fears bugs.'

Furthermore, Chinese EO verbs lexically specify if they can take an agent subject and therefore, not all EO psych verbs exhibit the kind of three-way ambiguity in (3). Transitive EO psych verbs can be categorized into three types based on whether they have an agentive use: those only allowing a non-agentive subject, those highly agentive hence only licensing an agent subject, and those licensing both types of subject. Non-exhaustive lists of verbs that fall into each of these three classes are shown below (see also Verhoeven 2017) .

(5) Three types of EO psych verbs

- I. Non-agentive EO verbs: *kùnrǎo* ('trouble, bother'); *jīngxià* ('frighten'); *zhènfèn* ('excite, encourage'); *zhènjīng* ('shock'); *yǐngxiǎng* ('affect');
- II. Agentive EO verbs: *qǐyuè* ('please'); *dǎrǎo* ('bother'); *kǒnghè* ('intimidate'); *dàimàn* ('neglect'); *sāorǎo* ('harass')
- III. Alternating EO verbs: *rěnnǎo/rěnnù* ('upset'); *gǔlì* ('encourage'); *màofàn* ('offend'); *ānwèi* ('comfort'); *zhémó* ('torture'); *fán* ('bother'); *cìjī* ('provoke, irritate'); *xiūrǔ* ('insult'); *jiàoyù* ('give...a lesson, teach'); *jīnù* ('anger'); *jīngdòng* ('shock, alert'); *tíxǐng* ('remind, warn'); *qǐfā* ('enlighten'); *shuǐfú* ('persuade'); *shāndòng* ('incite')

Notably, both agentive and non-agentive EO psych verbs can take animate subjects; to distinguish agentive subjects from non-agentive subjects, we may ask whether they can license agent-oriented adverbials such as *gùyì/cúnxīn/yǒuyì* 'intentionally' with an animate subject. The sentences in (6) show that purely non-agentive EO verbs are not compatible with the adverb even

when the subject is animate; the subject can only be interpreted as an animate non-agent, whose volition is not involved, e.g., (6a) is interpreted as ‘his existence or his story encouraged me’.

- (6) a. Tā (*gùyì) zhènfèn-le wǒ.
he intentionally encouraged-PFV me
‘His existence encouraged me.’
b. Tā (*gùyì) kùnrǎo-le wǒ.
he intentionally bother-PFV me
‘His existence bothered me.’

We focus on the agentive-non-agentive alternating EO psych verbs as they allow both an agent and a non-agentive subject (as diagnosed by agent-oriented adverbs when animate), thus serving as a suitable test ground for the agentive/non-agentive contrast and analyses of the sources of non-culmination in Mandarin.

Focusing only on the alternating EO psych verbs, we note another key difference regarding their semantic entailments: the non-agentive variant enforces that the change in mental state of the object must come to hold, whereas the agentive variant does not enforce this, allowing zero change in mental state. (7) illustrates this basic contrast with the verb *xiūrǔ* ‘humiliate’.

- (7) a. duìshǒu xiūrǔ-le Lǐsì, dàn Lǐsì yìdiǎn méi gǎnjué.
rival humiliate-PFV Lisi but Lisi a.bit NEG feel
Intended: ‘The rival humiliated Lisi, but he didn’t feel anything.’
b. duìshǒu-de shènglì xiūrǔ-le Lǐsì, #dàn Lǐsì yìdiǎn méi gǎnjué.
rival-DE victory humiliate-PFV Lisi but Lisi a.bit NEG feel
Intended: ‘His rival’s victory humiliated Lisi, but he didn’t feel anything.’

This effect is reminiscent of the agent and non-agent contrast in terms of culmination entailments observed with accomplishment predicates in Chinese, a domain much more well-studied in the literature (e.g., Martin et al. 2020, 2021). The following example illustrates this with the accomplishment predicate *guān* ‘close’.

- (8) a. Lùlù guān-le nèi-shàn mén, dàn méi guān-shàng.
Lulu close-PFV that-CL door but NEG close-up
Intended: ‘Lulu closed that door, but it didn’t get closed at all.’
b. Nà-zhèn fēng guān-le nèi-shàn mén, #dàn méi
that-CL wind close-PFV that-CL door but NEG
guān-shàng.
close-up
Intended: ‘The gust of wind closed that door, but it didn’t get closed at all.’

This basic contrast has additionally been observed in the accomplishment predicates of a variety of unrelated languages, as illustrated below.

(9) *Indonesian*

- a. Esti men-tutup pintu, tapi tidak ter-tutup.
 Esti TR-close door but NEG INTR-close
Intended: 'Esti closed the door, but it didn't close.'
- b. Angin men-tutup pintu, #tapi tidak ter-tutup.
 wind TR-close door but NEG INTR-close
Intended: 'The wind closed the door, but it didn't close.'

(Sato 2019)

(10) *Korean*

- a. Chelswu-nun mwul-ul el-li-ess-una, mwul-i an
 Chelswu-TOP water-ACC freeze-CAU-PAST water-NOM
 an el-ess-ta.
 NEG freeze-PFV-DEC
Intended: 'Chelswu froze the water, but the water did not freeze.'
- b. Hanpa-ka gangmwul-ul el-li-ess-una #ganmwul-i an
 cold.wave-NOM river-ACC freeze-PAST-BUT river-NOM NEG
 el-ess-ta.
 freeze-PAST-DEC
Intended: 'A cold wave froze the river, but the river didn't freeze.'

(Martin 2020: 259-260)

Additionally, the same contrast is observed in EO psych verbs of various languages; an example from French is given below.²

- (11) a. *Pierre l'a provoquée, mais cela ne l'a pas*
 Pierre her has provoked but this NEG her has NEG
touchée du tout.
 touched at all
 'Pierre provoked her, but this didn't touch her at all.'
- b. *Cette remarque l'a provoquée, # mais cela ne l'a pas*
 this remark her has provoked but this NEG her has NEG
touchée du tout.
 touched at all
 'This remark provoked her, but this didn't touch her at all.'

(Martin and Schäfer 2017: 89)

One prominent account for non-culminating accomplishments in various languages focus on the semantic contribution of the perfective, arguing that the perfective marker in these languages, such as *-le* in Chinese, are weak perfective markers which can maximize any sub-part of an event,

² As noted by Martin (2020), in Romance and Germanic, the non-culmination effect persists only in certain verb classes like EO psych verbs and not in perfective accomplishments. Accounts for non-culmination in such languages therefore cannot be the same as that for languages that allow non-culmination with perfective accomplishments (Martin & Schäfer 2017; Martin 2020). We return to this in section 3.1.

Martin (2020) therefore suggests there is a fundamental difference between agentive and non-agentive uses of accomplishment predicates in terms of their event structure. Put simply, in a non-agentive use of an accomplishment predicate, there is *only one event participant* in the actual change-of-state event denoted by the VP, i.e., the theme whose state changes, while the non-agentive subject denotes another event that causes the change-of-state.⁴ Because the perfective marker applies to the VP, it maximizes parts of the VP event, which are by themselves changes-of-state, meaning that denial of any change will be infelicitous.

- The non-agentive use of an accomplishment contrasts with an agentive use; in the agentive use, the key intuition is that the subject is an agent of the actual change-of-state event and their contribution is in carrying out some action that causes the state of the window to be closed.⁵ Since the change-of-state consists of *two event participants*, the theme undergoing the change of state and the agent's action, the weak perfective marker either maximizes a complete change-of-state

(Martin 2020: 266)

⁵ Formally, this is implemented by Kratzer's agentive Voice head combining with the VP via Event Identification, such that the event the subject argument is an agent of is identified with the event denoted by the VP. In the non-agentive case, Martin (2020) proposes a different 'flavor' of Voice, which introduces a causing relation between the event denoted by the subject and the VP.

event (i.e., Lùlù-did-something-to-close-the-door) or a fragment of the event corresponding to the agent's action *prior to triggering a change-of-state of the theme* (i.e., Lùlù did something). Because of these two options, the agentive use of a perfective accomplishment is therefore predicted to be optionally culminating.

- (13) Lùlù guān-le nèi-shàn mén.
 Lùlù close-PFV that-CL door
 'Lùlù closed the door.'

≈Lulu's actions are part of a caused change-of-state of the door e causing the state of the window s to be closed.

Against this backdrop, we may now return to the core contrast in EO psych verbs illustrated in (7). Since Chinese accomplishment predicates exhibit the same effect as in (8), a natural account would be to assimilate EO psych verbs with accomplishment predicates in terms of their event structure across agentive and non-agentive uses, thereby naturally predicting them to pattern together in terms of non-culmination. For such a unified account, two core assumptions would need to be made about EO psych verbs and the distribution of non-culminating readings:

- First, the distribution of non-culminating readings must partly be due to the semantics of the perfective marker *-le*, as with regular accomplishment predicates.
- Second, the event structures of EO psych verbs across agentive and non-agentive uses must share at least a common change-of-state core, consisting of an event of change and a result state as demonstrated in Martin's (2020) analysis of accomplishment predicates illustrated in (12)-(13).

Should these two assumptions be untenable, then a unified account of non-culmination across accomplishment predicates and EO psych verbs would become impossible. In what follows, we demonstrate that this is exactly what is observed with EO psych verbs: the non-culminating effect in the mental change-of-state denoted by these verbs is *independent of perfective marking*, and that a range of diagnostics demonstrate that the agentive and non-agentive use of these verbs systematically differ in their event structures such that some uses do not in fact denote any change of state. Taken together, this means that an account for their non-culminating readings must be sought elsewhere instead of perfective marking and their event structures.

3. A closer look at Mandarin EO psych verbs

In this section, we aim to demonstrate two issues that determine a correct analysis of the distribution of non-culminating readings. First, non-culmination of the change-in-mental state denoted by Chinese EO psych verbs is independent of perfective marking. Second, the event structures of the agentive and non-agentive uses are not uniform, as already hinted at by Landau (2010) and Temme (2018). Taken together, this will lead us to reject an analysis based on weak perfectivity and event structure as an adequate account of non-culmination in Chinese EO psych verbs.

3.1 The non-influence of perfectivity

It has been argued that perfectivity is not the key factor for non-culmination in accomplishments in Mandarin (Lin 2004; Koh and Soh 2005; Koenig and Chief 2008). In this section, we give further evidence to show that if at least part of what leads to non-culminating readings is contributed by a weak perfective marker in Chinese, then a natural expectation would be that in the absence of perfective marking, non-culminating readings would either be absent, or that we would not expect any contrast between agentive and non-agentive uses of EO psych verbs and indeed, of accomplishment predicates in general. Martin (2020) and Martin et al. (2021) do not address this prediction explicitly; we note, however, that the agentive and non-agentive contrast persists even in explicitly *imperfect contexts with accomplishment predicates* in Chinese. This is notable, since one might expect non-culmination to be the default reading with the imperfective as its intuitive meaning is precisely that an event is ongoing and makes no reference to the culmination. Nonetheless, native speakers observe the same basic contrast as with perfective uses of accomplishments.

- (14) a. Zhāngsān zhèngzài shāo shū, dàn shū méi shāo-zháo.
Zhangsan PROG burn book but book NEG burn-RES.
Intended: 'Zhangsan was burning the book, but it didn't get burnt.'
- b. Nà bǎ huǒ zhèngzài shāo shū, #dàn shū méi shāo-zháo.
that CL fire PROG burn book but book NEG burn-RES
Intended: 'That fire was burning the book, but it didn't get burnt.'

Focusing here on Chinese EO psych verbs, we observe the same contrast likewise persists with the imperfective, as shown in (15) and (16).

- (15) a. Lùyl zhèngzài cǐjī tā, dàn tā yìdiǎn méi juéde.
Louis PROG trigger he but he a.bit NEG feel
Intended: 'Louis was triggering him, but he didn't feel a bit.'
- b. Zhè piān bàodào zhèngzài cǐjī tā, #dàn tā yìdiǎn
this CL report PROG trigger he but he a.bit
méi juéde.
NEG feel
Intended: 'This report was triggering him, but he didn't feel a bit.'
- (16) a. Lǎoshī zhèngzài gǔlì tā, dàn tā yìdiǎn méi juéde.
teacher PROG encourage he but he a.bit NEG feel
Intended: 'The teacher was encouraging him, but he didn't feel a bit.'
- b. Zhè piān bàodào zhèngzài gǔlì tā, #dàn tā
this CL report PROG encourage he but he
yìdiǎn méi juéde.
a.bit NEG feel
Intended: 'This report was encouraging, but he didn't feel a bit.'

This is, in fact, not an isolated fact observed only in Chinese EO psych verbs; perfectivity is also not required for the contrast to arise with respect to non-culmination in some Romance and

Germanic languages that generally do not allow non-culmination in other non-psych accomplishments (Martin & Schäfer 2012, 2017; Martin 2020). (17) illustrates with an example from French, where the contrast is shown even in the *future* aspect.

- (17) a. *Pierre lui enseignera le russe, mais elle ne l'apprendra pas.*
 Pierre her teach.FUT the russian but she NEG
 it learn.FUT NEG
Intended: 'Pierre will teach her Russian, but she won't learn it.'
- b. *Ce voyage lui enseignera le russe, #mais elle ne l'apprendra pas.*
 this trip her teach the russian but she NEG
 it learn.FUT NEG
Intended: 'This trip will teach him Russian, but he won't learn it.'
- (Martin & Schäfer 2017: 94-95)

Since Romance and Germanic languages like French and German do not generally allow non-culmination with accomplishment predicates in the perfective and the perfective is not present in examples like (17), Martin and Schäfer (2012) assume that non-culmination in this class of verbs (what they call *defeasible causatives*) have a different source from that observed in perfective accomplishment predicates. The key intuition is that EO psych verbs in languages like French contain a shared bi-eventive change-of-state core, which are crucially under the scope of a sub-lexical modal operator. The contrast between agents and non-agents falls out as a choice of the modal base of the sub-lexical operator: agentive subjects enforce an energetic modal base containing worlds where the goals of the agent are achieved (which do not necessarily contain the evaluation world), whereas non-agentive subjects impose a circumstantial modal base, which necessarily quantifies over the world of evaluation and therefore the change-of-state must hold (see also Martin 2015; Martin and Demidarche 2020).

We may now once again ask if such an account will extend to Chinese EO psych verbs, given that perfectivity is not required for the non-culmination contrast. This will again rest on whether one can analyze the event structures of the agentive and non-agentive uses as sharing a bi-eventive change-of-state core, above which one may postulate a sub-lexical modal operator. We now proceed to show, using standard diagnostics, that the event structures of the agentive and non-agentive uses of Chinese EO psych verbs are, in fact, not uniform and fall at least into two distinct groups, depending on how one interprets the results of the various diagnostics.

3.2 Event structure of Chinese EO psych verbs

Identifying the event structure of EO psych verbs remains a highly controversial endeavor. While some suggest that most EO psych verbs are stative, with a small number of them having an inchoative reading (Dowty 1991), some argue that EO verbs are all inchoative, and are hence achievement verbs (Van Voorst 1992). Martin & Schäfer (2017: 96-98) give three pieces of

evidence based on event structure tests⁶ to show that EO verbs share a bi-eventive change-of-state core across agentive and non-agentive uses in German: i) the availability of restitutive readings with *again*-modification, ii) compatibility with time-frame adverbials that measure the process portion of the change of state, and iii) compatibility with durative adverbials that measure how long a reversible result state holds.

Beginning with restitutive readings, (18) shows that *the state of me being encouraged without previous causation* can be restored. That is, the sentences in (18) can be judged true if I were always an inherently motivated person but went through a period of self-doubt before this conversation / Hans managed to encourage me again. Under a view of the adverb *again* as a sub-lexical modifier which can attach to a result state sub-constituent independent of the sub-event denoting the process (e.g., Dowty 1979; von Stechow 1996, a.m.o.), this means that the verb *encourage* in German has a bi-eventive change-of-state event structure regardless of the properties of the subject.

- (18) a. *Dieses Gespräch hat mich endlich wieder ermutigt.*
 this conversation has me finally again encouraged
 'Finally, this conversation encouraged me again.'
- b. *Hans hat mich mit seiner Rede endlich wieder ermutigt.*
 Hans has me with his discourse finally again encouraged
 'Finally, Hans encouraged me again with his talk.'
- (Martin & Schäfer 2017: 96)

Second, in a bi-eventive change-of-state event, time-frame adverbials can measure the duration of the process at whose culmination the change-of-state occurs. This is typically described with an *in x-time* phrase in English; Martin & Schäfer (2017: 96) demonstrate that EO psych verbs like *reassure* in French are indeed compatible with an *in-phrase* describing the duration of the process.

- (19) *Pierre l'a rassurée avec ses mots doux en cinq minutes.*
 Pierre her-has reassured with his sweet words in five minutes
 'Pierre reassured her with his sweet words in five minutes.'

Thirdly, the agentive use of a psych verb, just like its non-agentive counterpart, is also compatible with durative adverbials that measure how long a reversible result state holds.

⁶ In addition to the three tests, they also provide a language-specific piece of evidence that comes from the fact that defeasible causative psych verbs, whether with agentive or non-agentive subject, can always form German *-ung* nominalization, as exemplified in (i), which can only be formed from bi-eventive resultative verbs.

- (i) *die Ermutigung der Kinder durch den Lehrer/ durch das Ereignis*
 the encouragement-of-the children by the teacher by the event
 'the encouragement of the children by the teacher/by the event'

- (20) *Hans ermutigte ihn einige Minuten entlang, aber dann*
 Hans encouraged him some minutes long but then
 verlor er seinen Mut wieder.
 lost his courage again
 ‘Hans encouraged him for some minutes, but then he lost his courage again.’
 (Martin & Schäfer 2017: 96)

We now apply the same three tests, demonstrating that for some EO psych verbs, the non-agentive variants show systemic event structural differences from their agentive counterparts.^{7 8} The key implication is that some Chinese EO psych verbs *cannot be clearly identified as accomplishments across their agentive and non-agentive uses* and yet persistently show a contrast in terms of culmination. Consequently, an account of non-culmination built on the premise that predicates that show these effects are bi-eventive accomplishments, as in the sub-lexical modality analysis, cannot hold as a general explanation for the contrast in non-culmination observed with EO psych verbs across their agentive and non-agentive uses.

#Diagnostic 1: Compatibility with the time-span adverbial *zài...(x-time) nèi (in-phrase)*

Since *in-phrase* measures the duration of a process that leads to a result (Dowty 1979; Binnick 1991; *a.m.o.*), compatibility of *in-phrase* with an EO psych verb will indicate that the verb is bi-eventive with independent durative process and result sub-components (Martin & Schäfer 2017). On the other hand, activities and states do not have a process that leads to a result, rendering them incompatible with such an adverbial. Achievements, on the other hand, describe an instantaneous change; *in-phrase* modifying achievements returns an *after-reading*, which

⁷ A reviewer suggests more event structure tests, such as the ambiguity of *too*-presupposition (Spathas & Michelioudakis 2021), and the entailment of the verb when used intransitively (Martin et al. 2023). The logic of the *too*-adverb test is similar to the classic *again*-test in targeting a result state, as the licensing condition of a result stative presupposition of *too* in Greek requires a stative result (Spathas & Michelioudakis 2021). However, we are uncertain whether the counterpart of additive *too* in Mandarin, i.e., *ye* ‘too’, has the same distribution and semantics for it to be used as a proper test for result entailment. Thus, we will leave the *too*-test for future study and only present the more well-studied *again*-test. The second diagnostic suggested to us involves the intransitive use of the EO verbs, is unfortunately not applicable in Mandarin as most of the psych verbs we are concerned with do not alternate between transitive/causative and intransitive, e.g., *?*Lùyì cǐjī le* (Louis trigger-PFV) cannot mean ‘Louis got triggered’, and is only marginally acceptable with an object-dropping reading (i.e., Louis triggered someone’).

⁸ There are some agentive/non-agentive alternating EO verbs that always culminate, namely, RVCs such as *jīngdòng* ‘shock’, *rěnnù* ‘anger’, *rěnnǎo* ‘irritate’, etc., where the first verb transparently describes the manner component while the second component is typically an adjective describing the resulting state. With respect to all the diagnostics, these verbs pattern like achievement resultative verbs, which is why they always entail that the mental state denoted by the second verb morpheme V2 is reached. In the following, we will only use them as our baseline to show that the tests given are viable, but in the summary presented in 3.3, we will set aside those verbs as they are irrelevant to our discussion of possible sources for non-culminating reading of EO verbs. On the other hand, the EO verbs we focus on in the following, e.g., *zhémó* (bend-grind, ‘torture’), *āndìng* (calm-stabilize, ‘comfort’), *cǐjī* (poke-provoke ‘provoke’), *xiūrǔ* (shame-humiliate, ‘humiliate’), are mostly coordinative compounds, where the two verb components have similar meanings, as indicated by the morpheme-by-morpheme gloss above, and are not amenable to an RVC analysis.

measures the time it took for the instantaneous change to occur⁹. For example, the resultative-like psych verbs we mentioned in fn.7 that always culminate are only acceptable on the reading where an *in*-phrase, the Mandarin counterpart being *zài...(x-time) nèi* (e.g., Zhang 2018), measures the duration leading up to the instantaneous change-of-state described by the verb, hence being synonymous with *after*-adverbials. (21a) shows that verbs like *jīngdòng* ‘shock’ and *rěnnù* ‘anger’ are incompatible with the adverb *mànmàn-de* ‘gradually’ intended to describe the manner in which a process event proceeds. With the *in*-phrase, it can only mean that the boss was shocked (in a moment) after one day, but not that there was a durative event that lasted one day and led to being shocked, which is the canonical reading of an accomplishment with *in*-phrases (see also Landau 2010: 129 for English and Fritz-Huechante et al. 2020: 11 for Spanish).

- (21) a. #Zhè xiāoxī/#Yuēhàn zài yì-tiān nèi (*mànmàn) jīngdòng-le
 this news/John in one-day within gradually shock-PFV
 lǎobǎn.
 boss
Intended: ‘This news/John gradually shocked the boss in one day.’
 OK: ‘This news/John shocked the boss after a day.’
- b. #Zhè xiāoxī/#Yuēhàn zài yì-fēnzhōng nèi rěnnù-le lǎobǎn.
 this news/John in one-minute within anger-PFV boss
Intended: ‘This news/John angered the boss in one minute.’
 OK: ‘The news/John angered the boss after a minute.’

Applying the *in*-phrase test to non-culminating EO psych verbs, we find that they can be split into two groups with respect to their compatibility with the adverbial: those completely excluding *zài...(x-time) nèi*, and those compatible with it on a time-span reading just as canonical accomplishments. First, verbs like *zhémó* ‘torture’ and *xiūrǔ* ‘humiliate’ are simply incompatible with the time-span adverbial whether the subject is agentive or non-agentive.

- (22) a. *Bìngtòng/*tā-de qīzi zài sān-nián nèi zhémó-le tā.
 illness/ he-MOD wife in three-year within torture-PFV he
 ‘It took three years for the illness/his wife to torture him.’
- b. *tā-de shībài/*Dírén zài sān-tiān nèi xiūrǔ-le tā.
 he-DE failure/ enemy in three-day within humiliate-PFV he
 ‘It took three days for his failure/the enemy to humiliate him.’

As for verbs like *shuifú* ‘persuade’, *ānfǔ* ‘comfort’, and *āndìng* ‘calm’, both agentive and non-agentive variants are compatible with the time-frame adverbial modifying an incremental process with an endpoint, as indicated by the modification of the adverbial *mànmàn* ‘gradually’.

⁹ This reading is also known as *ingressive*, which is argued to be the result of a repair mechanism as the psych predicates lack a process leading to an endpoint, which could be measured by the adverbial. See Pinõn (1997) for a thorough discussion on achievements and Fritz-Huechante et al. (2020) for a discussion on the ingressive reading with psych verbs.

- (23) a. Cánkù-de xiànshí/tā zài sān-nián nèi mànman
harsh-MOD reality/he in three-year within gradually
shuǐfú-le Lǐsì (shēnghuó bùyì)
convinced-PFV Lisi life hard
'The harsh reality/he gradually convinced Lisi (that life is hard) in three years.'
- b. Érzy-degūilái/Érzy zài jǐ-tiān nèi mànman ānfǔ-le tā.
son-de return/son in several-day within gradually comfort-PFV he
'His son's return/his son gradually comforted him in a few days.'
- c. Xīn zhèngcè/Lǐngdǎo zài jǐ-tiān nèi mànman āndìng-le
new policy / boss in several-day within gradually calm-PFV
yuángōng.
employee
'The new policy/the boss gradually calmed the employees in a few days.'

Predictably, with the time frame phrases specifying the duration of the bounded process, the event must culminate even in the agentive use, as shown by the infelicity of the following denial of the result in (24); this therefore confirms that these predicates behave like bi-eventive accomplishments.

- (24) a. tā zài sān-tiān nèi shuǐfú-le Lǐsì, #dàn Lǐsì hái
he in three-day within persuade-PFV Lisi but Lisi still
méi bèi shuǐfú.
not PASS persuade
Intended: 'He persuaded Lisi in three days, but Lisi is still not persuaded.'
- b. Érzy zài jǐ-tiān nèi mànman ānfǔ-le tā, #dàn
son in several-day within gradually comfort-PFV he but
tā hái méi bèi ānfǔ.
he still NEG PASS comfort
Intended: 'His son gradually comforted him in a few days, but he's still not comforted.'

The result of this test shows that some EO psych verbs such as *zhémó* 'torture', *yāyī* 'depress' pattern as activities or states in that they are incompatible with the time-frame adverbial, while other verbs (e.g., *shuǐfú* 'convince/persuade') are on a par with canonical accomplishments, which entail a process leading to a change-of-state across their agentive and non-agentive uses.

#Diagnostic 2: result state-reading of durational adverbials

To further show that the non-culminating psych verbs are result-entailing accomplishments, Martin & Schäfer (2017) use the durational adverb *for*-phrase, measuring the time interval a reversible result state holds¹⁰. Specifically, if a verb is compatible with both an *in*-phrase and *for*-

¹⁰ Note that not all accomplishments have a result that can hold and be measured; e.g., *draw a picture* as an accomplishment does not allow a durational adverbial modifying the result state, i.e., the existence of the picture (Pinõn 1999), and hence *draw a picture for ten minutes* can only mean the picture-drawing process lasts ten minutes. However, such accomplishments are derived from activities by adding an

phrase specifying how long the result state holds, then we can be sure it behaves like a bi-eventive accomplishment containing two subevents, namely, the causing event and the result event.

With this in mind, we turn to the Chinese counterpart of *for*-phrases, which is expressed by *V-le-NP X-time*¹¹. Again, we first test the *jīngdòng*-type verbs, which we mentioned behave like RVCs without a duration and pattern like non-durative achievements. They are predictably only compatible with the adverb on a result state-reading (RS-reading)¹².

- (25) a. Zhè xiāoxī/Yuēhàn jīngdòng-le lǎobǎn yì-tiān.
 this news/John shock-PFV boss one-day
 'The boss has been shocked by the news/John for a day.' [RS-reading]
- b. Zhè xiāoxī/Yuēhàn rěnnù-le Lùyì yì-tiān.
 this news/John anger-PFV Louis one-day
 'Louis has been shocked by the news/John for a day.' [RS-reading]

Turning to the class of non-culminating EO psych verbs, the *zhémó*-group, which rejects the modification of the time-frame adverb (recall (22)), can license *V-le-NP X-time* constructions. Interestingly, with a non-agentive subject, the only available reading is the RS-reading, while with an agent as the subject, the eventuality-related reading (E-reading) becomes the only reading. The RS-reading in the non-agentive sentences can be further strengthened with the follow-up sentence describing the reversed result state, as shown in (27) (this can be done for the other verbs as well).

- (26) a. Qīzi-de sǐqù/Tā-de qīzi zhémó-le tā sān-nián.
 wife-DE death/he-DE wife torture-PFV he three-year
 'His wife's death tortured him for three years.' [non-agentive: RS-reading]
 'His wife spent three years torturing him.' [agentive: E-reading]
- b. Tā-de shībài/Dírén xiūrǔ-le tā sān-nián.
 he-DE failure/enemy insult-PFV he three-year
 'His failure has been insulting him for three years.' [non-agentive: RS-reading]
 'The enemy spent three years insulting me.' [agentive: E-reading]

incremental theme and hence might be different from change-of-state accomplishments, which are truly bi-eventive (see Rappaport 2014).

¹¹ The durational phrase can be ambiguous when modifying an event entailing both a process and a result state. For example, (i) can mean either the building event lasted three years, which we call eventuality-related reading (E-reading), or the existence of the house holds for three years, which we term result state-related reading (RS-reading) (following Pinõn 1999).

- (i) Fángzi zào-le sān-nián.
 house build-PFV three-year
 i. 'The house was under construction in the past three years.' [E-reading]
 ii. 'It has been three years since the house was built.' [RS-reading]

¹² It is worth pointing out that the sentences in (25) with the agent subject can still be compatible with the *for*-phrase on an E-reading, which, however, requires iterativity of the event, i.e., the event keeps repeating itself. For instance, (25a) with the agent subject is felicitous in the scenario where John repeatedly told his boss something shocking throughout the day. The iterative reading is common when an accomplishment or an achievement is modified by *for*-phrases (Pinõn 1999).

- c. Zhè-jàn shì/Yuēhàn cìjī-le wǒ sān-tiān.
 this-CL incident/teacher provoke-PFV I three-day
 'This incident provoked him for three days.' [non-agentive: RS-reading]
 'The teacher spent three days provoking me.' [agentive: E-reading]
- d. Zhè-cì huòshèng/Lǎoshī zhǐ gǔlì-le Lùyì sān-tiān.
 this-CL victory/teacher only inspire-PFV Louis three-day
 'This victory only encouraged Louis for three days.' [non-agentive: RS-reading]
 'The teacher spent three days encouraging Louis.' [agentive: E-reading]
- (27) Zhè-cì huòshèng zhǐ gǔlì-le Lùyì sān-tiān, sān-tiān
 this-CL victory/teacher only inspire-PFV Louis three-day three-day
 hòu tā yòu shīqù-le xìnxīn.
 after he again lose-PFV confidence
 'This victory only encouraged Louis for three days. After three days, he lost his confidence again.'

Put together with the result of the time-frame adverbial test, this indicates that the EO psych verbs of the *zhémó*-group do not pattern like bi-eventive accomplishments. The contrast observed between the agentive and non-agentive uses indicates that the agentive variant behaves like an activity, rejecting *in*-phrases and having only the E-reading with *for*-phrases, while the non-agentive variant patterns like a state, rejecting *in*-phrases and having only the RS-reading with *for*-phrases.

The other group of EO verbs include verbs such as *shuifù* 'convince/persuade', *āndìng* 'calm', etc. This group of verbs do not show a contrast between the agentive and the non-agentive uses: both disallow modification with the durational time adverbial with the RS-reading that the result state holds for a period of time until it was reversed (as illustrated by the follow-up statement).

- (28) a. *Tā de jiěshì/*Yuēhàn shuifú-le Lùyì sān-tiān,
 he DE explanation/John persuade-PFV Louis three-day
 sān-tiān hòu tā gǎibiàn-le zhǔyì.
 three-day after he change-PFV mind
 'His explanation/John persuaded him for three days, but he changed his mind after three days.' [RS-reading]
- b. *Xīn zhèngcè/*Lǐngdǎo āndìng-le yuángōng jǐ-tiān.
 new policy/boss calm-PFV employee several-day
 kě méiduōjiǔ tāmen yòu kāishǐ bùān le.
 but not.long they again start upset PFV
 'The new policy/the boss calmed the employees for a few days, but it was not long before they started being upset again.' [RS-reading]

The results for this group of EO verbs therefore consistently pattern like mono-eventive events, where there is no accessible result that can be modified by a *for*-phrase with a RS-reading.

#Diagnostic 3: *yòu* ‘again’-presupposition

It is well-established that in many languages, the adverb AGAIN induces presuppositional ambiguity between a repetitive reading and a restitutive reading when modifying accomplishments that have a result state (Dowty 1979; von Stechow 1996; Beck 2006; Xu 2016; Yu 2020). Hence, the *again*-presupposition test can help identify the subevents a predicate encodes. As shown by the scenarios given for the two readings in (29), the agent must have done the opening twice on the repetitive reading, and the restitutive reading only requires the door to be in the open state twice, without a previous causing event. The ambiguity arises from AGAIN targeting different subevents entailed by the predicate, i.e., a CAUSE event (actions of opening) and a BECOME event (the door being in an open state).

- (29) Zhāngsān yòu dǎ-kāi-le mén.
Zhangsan again open-open-PFV door
‘Zhangsan opened the door again.’

Scenario i: (*repetitive reading*) Zhangsan opened the door before, and after a while, someone else closed the door. Feeling hot, ...

Scenario ii: (*restitutive reading*) Lisi built a wardrobe for Zhangsan. He set the door on its hinges and then closed it. Curious about what was inside the wardrobe, ...

(Xu 2016: 24)

Following Martin & Schäfer (2017), we use *yòu* ‘again’-presupposition to test whether a psych verb entails a result arising from a process, i.e., whether there are two separate sub-events within the event structure¹³. As the repetitive reading is the default reading of AGAIN, the crucial test should focus on whether there is a restitutive reading (presupposition of a previous state to the exclusion of the causing event): the existence of the restitutive reading provides evidence for a bi-eventive structure. Looking at the *zhémó*-group, we find that only their non-agentive variant allows restitutive readings, as exemplified by (30)-(31). Note that in the context, the prior mental state of the referent of the surface object was not the result of any process event.

- (30) **Scenario:** John had always been confident in chess and working very hard on securing the championship, until he lost the most important game in his career because of a silly mistake and ever since he was not as confident as before.

¹³ Note that the *again*-presupposition test should be used with some caution. Many factors, such as the entailment of manner by the root can affect the possibility of the restitutive reading (see Beavers & Koontz-Garboden 2020 for a detailed discussion). For instance, for some psych verbs, such as *màofàn* ‘offend’ and *zhémó* ‘torture’, the mental state (e.g., ‘being offended’) just cannot exist without having a cause due to world knowledge, hence the absence of the restitutive reading whether the subject is an agent or not. What is crucial here is the contrast between the two variants with respect to restitutive readings - the existence of such a contrast will be evidence in support of our argument that EO verbs do not necessarily share a change-of-state core (also see Martin & Schäfer’s 2017).

- a. {His coach gave him a book about a man who never won in his life but never gave up. After reading the book, he was so touched and decided to be confident and work hard as before.}

Zhè běn shū yòu gǔli-le Yuēhàn.

this CL book again encourage-PFV John

'This book encouraged John again.'

- b. {His coach talked to him everyday and encouraged him to never give up. He was so touched and decided to be confident and work hard as before.}

??Jiàoliàn yòu gǔli-le Yuēhàn.

coach again encourage-PFV John

'The coach encouraged John again.'

(31) **Scenario:** John was born with low self-esteem. After years of confidence building, he has become more confident than before and even decided to run for mayor. His competitor wanted to take advantage of his struggle for self-esteem, so...

- a. {he spread the rumor that John was involved in a horrible scandal, which made him feel humiliated and started to have low self-esteem again.}

Zhè gè yáoyán yòu xiūrǔ-le Yuēhàn.

this CL scandal again humiliate-PFV John

'The scandal humiliated John again.'

- b. {he deliberately made fun of John in public, which made him humiliated and started to have low self-esteem again.}

??Yuēhàn-de duìshǒu yòu xiūrǔ-le tā.

John-DE competitor again humiliate-PFV he

'John's competitor humiliated him again.'

On the contrary, it is observed that the second group of EO verbs allows the restitutive reading both with an agent subject and a non-agent, such as *āndìng* 'calm' and *shuifú* 'persuade'. As shown in (32), whether used agentively or non-agentively, *āndìng* 'calm' can describe the scenario where the causing event only happened once, but the referent of the object experiences the mental state twice, i.e., the restitutive reading.

(32) **Scenario:** The employees had been peacefully working in this small company for many years, but one day rumor came that the company was in deep debt and was going to fire at least half employees to cut down the cost. The employees all panicked until...

- a. {the company issued a new policy that there will be no one fired. The employees were relieved and went back to daily work as before.}

Xīn zhèngcè yòu āndìng-le yuángōng.

new policy again calm-PFV employee

'New policy calmed the employees again.'

- b. {the boss gave a speech and promised that no one will be fired. The employees were relieved and went back to daily work as before.}
 Lǎobǎn yòu āndìng-le yuángōng.
 boss again calm-PFV employee
 'The boss calmed the employees again.'

(33) **Scenario:** John used to believe that the earth is round, but since he joined a religious cult which brainwashes its followers into believing that the earth is flat, he has lost his common sense, until one day...

- a. {he saw a picture of the earth taken from outerspace by NASA, which changed his mind that made him believe again that the earth is indeed round.}
 Zhè-zhāng zhàopiàn yòu shuìfú-le Yuēhàn dìqiú shì yuán-de.
 this-CL picture again persuade-PFV John earth BE round-DE
 'This mistake persuaded John again that the earth is round.'
- b. {his teacher lectured for days and persuaded him that the earth is round.}
 Lǎoshī yòu shuìfú-le Yuēhàn dìqiú shì yuán-de.
 teacher again persuade-PFV John earth BE round-DE
 'The teacher persuaded John again that the earth is round.'

If *yòu*-ambiguity arises from interactions with either the process or the result sub-events as widely assumed, the contrast observed from (30) and (31) would indicate that the agentive use of the first class of EO verbs does not have a result state to the exclusion of a causing event, corroborated further by the fact that they reject *in*-phrases and produce E-readings with *for*-phrases. These suggest that they behave like activities and have a mono-eventive structure. On the other hand, the non-agentive use with *yòu* is licensed without a prior repetitive context. Together with the fact that the non-agentive use of these verbs also reject *in*-phrases and only have RS-readings with *for*-phrases, we might conclude that they are stative and in (30a) and (31a), *yòu* is in effect attaching to a stative event and its presupposition is satisfied by the prior state.

As for the *shuìfú*-group, both the agentive and non-agentive uses license restitutive *yòu*; *prima facie*, this seems to be inconsistent with their incompatibility with the *for*-phrase indicating a mono-eventive structure. At present, it is not clear therefore what event structure the *shuìfú*-group of EO psych verbs instantiate; as noted by Zhu et al. (2025) who replicated these tests across a larger number of native speakers, there is significant variation in terms of whether a *for*-phrase is licensed with RS-readings. Crucially, regardless of whether a particular speaker accepts a RS-reading with *for*-phrases, the contrast between agents and non-agents with respect to culminating readings nonetheless persists *across all speakers*. This therefore strongly suggests that the distribution of non-culminating readings does not correlate with event structure, as we have been alluding to here, and this is further corroborated with the *zhémó*-group behaving differently with respect to the diagnostics across the agent and non-agentive uses and still showing the same contrast with non-culmination. We will therefore not pursue the question of what exactly the event structural properties of both groups of EO verbs are in further detail;

instead, we pursue an analysis that is independent on event structure and bi-eventivity, which we turn to in section 4.

3.3 Interim summary

Applying the above diagnostics, it is therefore clear that EO psych verbs do not always share the same *aktionsart* and event structure properties, as is most evident for the *zhémó*-type verbs which crucially alternate between activities and states across agentive and non-agentive uses, and neither variant of which entails a change of state sub-event at all. We therefore confirm previous observations that agentive and non-agentive variants of EO psych verbs do not share the same event structure (also see Pestesky 1995; Klein & Kutscher 2005; Cheung & Larson 2015; Temme 2018 for the same conclusion). For ease of exposition, we summarize the results of the above diagnostics as follows with representative verbs.

EO verbs	time-span adverb		durational adverb (RS-reading)		<i>again</i> -presupposition (restitutive reading)	
	Non-agent	Agent	Non-agent	Agent	Non-agent	Agent
<i>zhémó</i> 'torture'	x	x	✓	x	n.a.	n.a.
<i>xiūrǔ</i> 'humiliate'	x	x	✓	x	✓	x
<i>cǐjī</i> 'provoke'	x	x	✓	x	✓	x
<i>gǔlì</i> 'encourage'	x	x	✓	x	✓	x
<i>shuìfú</i> 'persuade'	✓	✓	x	x	✓	✓
<i>ānfǔ</i> 'comfort'	✓	✓	x	x	✓	✓
<i>āndìng</i> 'calm'	✓	✓	x	x	✓	✓

Table 2 Aspectual properties of EO verbs

(✓=compatible with the intended reading, x=incompatible with the intended reading, n.a.=nonapplicable)

To summarize, Chinese EO psych verbs that allow non-culmination in the agentive use are heterogeneous in terms of their behavior and whether they behave like bi-eventive accomplishments with process and result sub-events. Together with the observation that the availability of non-culminating readings with EO psych verbs is also not dependent on perfective marking, a maximally satisfactory explanation of the agent non-agent contrast with respect to non-culmination should therefore lie elsewhere other than the analyses currently available in the literature, namely i) in weak perfectivity encoded by the perfective marker *-le* and the availability of a change of state event sub-event in their event structures or ii) the presence of a sub-lexical modal operator with a bi-eventive change of state structure. We devote the next section to locating the source of the contrast in EO psych verbs.

4. The source of the agentive/non-agentive contrast in EO psych verbs

In this section, we argue that the source of the agentive/non-agentive contrast in EO psych verbs lies in the different thematic interpretations of the objects in the agentive use and the non-agentive use, rather than the presence of perfective marking and the assumption that EO psych verbs share a change of state sub-event core. The key intuition is that with agentive subjects, the object is interpreted as a theme and the mental state of the referent of the surface object is not *necessarily* implicated, allowing for non-culmination of the change in mental state. With non-agentive subjects, the surface object is interpreted instead as an experiencer, with the consequent entailment that the referent's mental state must be implicated, ruling out non-culminating readings. Building on Cheung & Larson (2015), we present evidence from the possibility of the surface object binding a structurally high reflexive anaphor contained in the subject across agentive and non-agentive subject constructions, contrasting with existing accounts in the literature regarding the source of non-culminating construals of accomplishment predicates in general.

4.1 Agentive vs. non-agentive EO verbs: binding asymmetries

It has been widely noted that EO psych verbs allow backward binding where the subject is bound by a non-c-commanding structurally lower DP, i.e., the object, in violation of the c-commanding constraint on anaphoric binding (Belletti & Rizzi 1988; Pesetsky 1995; Reinhart 2001). This is illustrated in English in (34a), in contrast to standard transitive sentences, as shown in (34b).

- (34) a. Rumors about himself_i enraged John_i. [EO verb]
 b. *Stories about himself_i described John_i accurately. [transitive verb]
 (Pesetsky 1995: 43)

Mandarin EO psych verbs also exhibit parallel anomalies in binding, as illustrated in (35).

- (35) a. Zìjǐ_i-de péngyǒu de guānhuái gǎndòng-le Lǐsì_i.
 self-DE friend DE solicitude touch-PFV Lisi
 'The solicitude of self_i's friends touched Lisi_i.'
 [EO verbs]

- b. *Zǐjǐ-de pényǒu pīpíng-le Lǐsì.
self-DE friend criticize-PFV Lisi
Intended: ‘Lisi’s friend criticized him.’ [simple transitive]
(Cheung & Larson 2015: ex. (9) & (11))

Based on the binding facts, Cheung & Larson (2015), recasting Belletti & Rizzi's (1988) analysis of Italian EO psych verbs in modern terms, argue that the surface subject in (35a) is base-generated within VP below the experiencer object. They further argue that EO verbs are systematically ambiguous between the two structural representations in (36), namely, agentive EO verbs and non-agentive EO verbs, the former involving a deep agent subject and a deep theme object, a run-of-the-mill transitive structure (see also Landau 2010; Kim 2024 for identical analyses in various languages), while the latter a double unaccusative structure, where both the experiencer and the source¹⁴ are generated within VP and the source raises to the subject position, à la Belletti & Rizzi (1988). The two structures and the thematic structures involved are given below¹⁵.

- (36) a.
- | | AGENT | | THEME | |
|--|-------------------------|---------------|--------------|--|
| | Mǎlì | jīnù-le | Lǐsi | |
| | Mary | infuriate-PFV | Lisi | |
| | 'Mary infuriated Lisi.' | | | |
- b.
- | | | EXP | SOURCE | | | |
|--|-------|---------------|---------------|-----------|----------|--------|
| | _____ | jīnù-le | Lǐsi | Mǎlì/Mǎlì | tūrán | líkāi. |
| | | infuriate-PFV | Lisi | Mary/Mary | suddenly | left |
| | ↑ | | | | | |

Evidence comes from the fact that an agent-oriented adverb, e.g., *gùyì* 'intentionally', blocks backward binding, as illustrated in (37), which indicates that the subject of an agentive EO verb is a deep subject, hence a normal transitive structure.

¹⁴ As pointed out by an anonymous reviewer, the surface subject argument in (36b) is often analyzed as a *causer* (e.g., Pesetsky (1995)), which explains why these cases are often paraphrased as *Mary's sudden departure caused Lisi to be infuriated*. However, we disagree with this analysis for two reasons. First, as we emphasized, not all non-agent uses of EO psych verbs here can be shown to be bi-eventive, which is implied by an analysis positing a causer since it requires a causative structure. Second, as already noted by Reinhart (2002) for English, backward binding into an anaphor in surface subjects of English EO verbs is possible, which she analyzes as stimulus rather than causer arguments. We elaborate on this point in 4.2.4, showing that Mandarin *ziji* is also sensitive to a thematic hierarchy in requiring its antecedent to be thematically higher than it in a thematic hierarchy. If the surface inanimate subject were a causer and therefore higher on the hierarchy than the surface object interpreted as experiencer, backward binding should not be possible, which is not borne out. We therefore do not adopt a causative analysis of the non-agentive use of EO psych verbs in Mandarin.

¹⁵ Note that not all analyses agree that the backward binding facts are due to the fact that the surface object is structurally higher than the surface subject (Landau 2010; Kim 2024). Nonetheless, the clear contrasts we will show below involving agentive adverbs blocking backward binding as well as the ditransitive cases suggests a clear difference in the status of the surface object (as also observed by Reinhart 2002; Sato and Kishita 2009) and we will continue to take it as reflective of a difference in thematic structure.

- (37) [Zǐjǐ-dedùishǒu] (?*gùyì) jīnù-le Lǐsì.
 self-DE opponent intentionally infuriate-PFV Lisi
 ‘Self_i’s opponent infuriated Lisi_i.’
 (Cheung & Larson 2015: ex. (124a))

So far, it is clear that agentivity plays a vital role in determining the underlying structure of a psych verb. We therefore propose that the source of the agentive/non-agentive contrast found in psych verbs with regard to culmination, which cannot be explained by either the weak perfectivity analysis (given that the same effects are found with imperfectives), or the sublexical modality/event structural hypothesis (as psych verbs that show the contrast are not necessarily uniformly accomplishments and bi-eventive, a prerequisite for these analyses), lies in the different thematic structures of agentive vs non-agentive use of EO verbs. When the subject is non-agentive, the object must be an *experiencer* who undergoes a change of mental state, while when the subject is an agent, the object is simply a theme, whose mental state is not obligatorily involved (see also Kim 2024 for recent views). To capture the difference formally, we adopt Reinhart’s (2002) θ -feature approach, where traditional θ -roles are specified with feature bundles made up of two binary features: $+/-c = \text{Cause change}$ and $+/-m = \text{Mental state}$ ¹⁶. What is crucial here is that experiencers are hence specified as $[-c+m]$, which means the mental state must be entailed by the verb’s thematic structure. Themes/patients, on the other hand, are specified as $[-c-m]$, contrasting with experiencers only in the $+/-m$ feature. Finally, the subject in the non-agentive use is assumed to be a stimulus/subject matter (in Reinhart’s terms), hence $[-m]$ (unary feature means the other feature, namely, $+/-c$, is unspecified and hence irrelevant to the lexical semantics); given the $-m$ specification, we expect these arguments to be base-generated in a structurally low position, namely within VP but lower than the experiencer argument (Reinhart 2002).¹⁷

¹⁶ The feature clusters can be defined as follows, but the correspondence of them to traditional θ -roles is not perfectly aligned.

- (i) $[+c+m]$ - agent; $[+c-m]$ - instrument (...); $[-c+m]$ - experiencer; $[-c-m]$ - theme/patient;
 $[+c]$ - cause (unspecified for $/m$; consistent with either agent or instrument)
 $[+m]$ - ? (can be the subject of verbs such as *love*, *hate*)
 $[-m]$ - (Unspecified for $/c$): **subject matter / locative source**
 $[-c]$ - internal roles like **goal**, **benefactor** typically dative (or PP)
 (Reinhart 2002: 232)

The system allows single feature clusters, where the other missing feature is irrelevant, hence allowing more freedom in contextual interpretations. For the sake of space, we refrain from going into further details. What is crucially relevant to our analysis is that experiencer has the feature $+m$ ($/\alpha$ is notated as a feature α) in its θ -feature specification.

¹⁷ According to Reinhart’s (2002: 246) mapping rules, a $[-]$ cluster (i.e., $[-c]$, $[-c-m]$, $[-m]$) is merged internally. We will make only these syntactic assumptions, namely, the relative structural heights of the agent, theme, experiencer, and stimulus arguments with respect to each other, and will not attempt to map the structural heights of the arguments to any particular event structure (mono-eventive or bi-eventive causative) due to the issues we noted at the end of section 3.2.

4.2 Predictions

If we understand experiencer as indicating that an individual's mental state is necessarily involved in the event as in Reinhart (2002), then the culminating reading of EO verbs with a stimulus/subject matter subjects falls out since the surface object argument's θ -features requires that the mental state of the argument be involved. With agentive subjects, the object is interpreted as a theme, requiring no change in mental state to be involved for the referent of the surface object. Based on these assumptions, we can make four predictions. We discuss these in turn in what follows.

4.2.1 Backward binding predicts non-culmination

If backward binding is possible in an EO verb sentence, the surface object should be an experiencer with a change of mental state, and therefore, the EO verb must not allow non-culmination, and vice versa. This prediction is borne out by (38). Backward binding is licensed in (38a), and as predicted, the non-culminating reading is impossible; the fact that the sentence is compatible with the given scenario shows that the subject, though animate, is interpreted as a non-agentive stimulus, therefore in line with our analysis. On the contrary, the scenario given in (38b) specifies the subject as an agent, and predictably, backward binding in this case is ruled out, and non-culmination becomes acceptable.

- (38) a. **Scenario:** Lisi had been in ranked 3rd in the world in chess for years and had repeatedly beat the player ranked 4th in the last year and therefore did not pay much attention to the skill level of this player. However, this 4th placed rival beat the top ranked player in a recent competition, surprising Lisi immensely. His rival's win over the champion made him realize that he should never underestimate any opponent.

Zìjǐ-de	dùishǒu	tíxǐng-le/zhe	Lǐsì	bùyào
self-DE	rival	remind-PFV/IMPFV	Lisi	do.not
qīngdǐ,	#dàn	Lǐsì	yìdiǎn méi	yìshí.
underestimate.enemy	but	Lisi	at.all	NEG realize

Intended: 'His rival reminded Lisi not to underestimate the enemy, but Lisi still didn't take it to heart.'

- b. **Scenario:** Lisi is not paying much effort in a chess game with Mary, and thus is about to lose, and Mary was not happy about being taken lightly so she deliberately told Lisi not to underestimate her.

Mǎlì /??Zìjǐ-de	dùishǒu	gùyì	tíxǐng-le	Lǐsì	bùyào
Mary self-DE	rival	deliberately	remind-PFV	Lisi	do.not
qīngdǐ,	dàn	Lǐsì	yìdiǎn méi	yìshí.	
underestimate.enemy	but	Lisi	at.all	NEG	realize

Intended: 'Mary/His rival deliberately reminded Lisi not to underestimate her, but Lisi still didn't take it to heart.'

4.2.2 Non-dependence of non-culmination on aspectual marking

Secondly, as θ -roles are bundles of lexical semantic information (whether specified in the lexicon or given by the structure) that are independent of the aspectuality of the event denoted by the verb, the current analysis, which only hinges on thematic features, can thus predict that whether in the perfective or imperfective, the argument that bears the feature $/+m$ should always require culmination in the change of mental state. We have already shown with the verb *xiūrǔ* ‘humiliate’ (repeated below), where the contrast is found in the agent vs non-agentive sentence, even in the progressive aspect.

- (39) a. Duìshǒu xiūrǔ-zhe Lǐsì, dàn Lǐsì yìdiǎn méi gǎnjué.
rival humiliate-IMP Lisi but Lisi a.bit NEG feel
‘The rival was humiliating Lisi, but he didn’t feel anything.’
b. Duìshǒu yíng-le de xiāoxī xiūrǔ-zhe Lǐsì, #dàn Lǐsì
rival win-PFV DE news humiliate-IMP Lisi but Lisi
yìdiǎn méi gǎnjué.
a.bit NEG feel
‘The news that his rival won humiliated Lisi, but he didn’t feel anything.’

4.2.3. EO verbs that always culminate and backward binding

If the crucial factor for EO psych verbs showing the agentive/non-agentive contrast is in the thematic interpretation of the object as an experiencer ($/+m$) such that the non-agentive case involves an experiencer object while the agentive case involves a theme object, then we predict that any EO psych verb that *always* lexically specifies its object as an experiencer will never allow non-culminating readings, regardless of whether its subject is interpreted agentively or as a stimulus/subject matter. One such verb is *gǎndòng* ‘touch/move (emotively)’. (40) shows the verb with an *inanimate* subject with an embedded anaphor. The inanimate subject must be interpreted as a stimulus/subject matter since it cannot be an intentional agent and hence generated low within the VP, allowing for backward binding. Since its surface object is lexically specified as an experiencer, a non-culminating reading is ruled out, on a par with the EO psych verbs we have examined so far. However, different from the other EO psych verbs we have seen, whose surface object’s thematic interpretation can alternate between theme and experiencer (depending on the theta-role of the subject), the object is always an experiencer even in its agentive use in (40b) (as confirmed by the adverb *gùyì* ‘on purpose’), and therefore continues to lack non-culminating readings. In that case, since the subject is an agent, backward binding becomes impossible. The prediction can be confirmed by (40c). This is crucial in understanding the relationship between non-culmination and backward binding: while backward binding indicates an unaccusative structure, where the surface subject (the stimulus) is generated lower than the surface object (the experiencer), whose theta-role specification imposes culmination, it is not the case vice versa: an object experiencer does not ensure the possibility of backward binding, because whether backward binding is allowed depends on the hierarchical relation between the subject and the object, not the theta role of the object alone.

- (40) a. Zìjǐ mǔqīn-de gùshì gǎndòng-le Lǐsì, #dàn Lǐsì yìdiǎn yě
 self mother-DE story touch-PFV Lisi but Lisi at.all still
 méi dònggróng.
 NEG moved
Intended: 'His_i mother's story touched Lisi_i, but he was not moved at all.'
- b. Mǎlì gùyì gǎndòng-le Lǐsì, #dàn Lǐsì yìdiǎn yě méi
 Mary intentionally touch-PFV Lisi but Lisi at.all still NEG
 dònggróng.
 moved
Intended: 'Mary touched Lisi, but Lisi was not moved at all.'
 (**Scenario:** Mary is deeply in love with Lisi, so she volunteered to help him with his work and take care of his pet in order to move him.)
- c. *Zìjǐ-de dùishǒu gùyì gǎndòng-le Lǐsì, (#dàn
 self-DE rival intentionally touch-PFV Lisi but
 Lǐsì yìdiǎn yě méi moved).
 Lisi at.all still NEG dònggróng.
Intended: 'His rival touched Lisi, but Lisi was not moved at all.'

4.2.4 Thematic prominence asymmetry of the surface (in)direct object

Last but not least, our analysis would predict that the object in the agentive variant of an EO psych verb to be thematically *less prominent* than the experiencer object in the non-agentive counterpart. The evidence that this is indeed true comes from (canonical) binding of the object into the complement clause of a ditransitive EO psych verb. As shown in (41a), when the subject *Zhāngsān* is a volitional agent, it is the preferred binder of the anaphor in the embedded clause, whereas the indirect object *Lǐsì* is unable to bind the embedded anaphor. Conversely, when the subject is non-agentive (e.g., intended to be interpreted as *that rival's abilities*), the same indirect argument *Lǐsì* is now *able* to bind the embedded anaphor *zìjǐ* even though the matrix animate subject is also a potential binder, which agrees with the anaphor both in animacy and number features.

- (41) a. Zhāngsān_i gàosù Lǐsì_i [zìjǐ_i hái bù gòugé].
 Zhangsan tell Lisi self yet NEG qualified
 'Zhangsan_i told Lisi_i that he_i is not qualified yet.'
- b. Nà gè dùishǒu gàosù Lǐsì_i [zìjǐ_i hái bù gòugé].
 that CL rival tell Lisi self yet NEG qualified
 'That rival's abilities told Lisi_i that he_i is not qualified yet.'
- (adapted from Xu 1994: ex. (13))

It is well-known that the anaphor *zìjǐ* in Chinese allows long-distance binding and is sensitive to the thematic prominence of the binder¹⁸ (Huang & Tang 1991; Xue & Pollard & Sag 1994; Xu

¹⁸ It is often the case that *zìjǐ* is subject-oriented; however, Xu (1994) points out that sometimes the object can be the binder when it is prominent on the θ -hierarchy.

1994; Pan 1995, 1997, 2001; Hu 2019; *a.m.o.*). Couched in a generally accepted thematic hierarchy, where agents are ranked highest with experiencers following it, followed by other thematic roles such as theme (Bresnan & Kanerva 1989; van Valin 1990; Grimshaw 1990), as shown in (42), Xu argues that a DP with a more prominent θ -role is superior when it comes to binding a non-c-commanding anaphor.

(42) Thematic Hierarchy

agent > experiencer > source/goal/locative > theme (Grimshaw 1990: 8)

Pan (1995, 1997, 2001) argues that the long-distance *zìjǐ* (but not the one that requires a local antecedent), which he calls the self-ascription *zìjǐ*, can be bound to a carrier of belief, the most prominent self-ascriber in the domain (Pan 2001: 297), and further, in the same spirit as Xu (1994), prominence is defined based on thematic hierarchy or syntactic structure. These analyses that hinge on *zìjǐ*'s sensitivity to thematic prominence can predict the contrast shown in (41). In (41a), the agent subject is the argument highest on the thematic hierarchy outranking the indirect object and therefore blocks the indirect object from binding the anaphor. In contrast to (41a), in (41b), even though the subject argument is an animate NP, as long as the context makes clear that it is a non-volitional source/stimulus argument (e.g., the rival's superior abilities made Lisi realize that he is not yet qualified), the indirect object can bind the anaphor, suggesting that it must outrank the subject argument in terms of the thematic hierarchy. We can therefore conclude that the indirect object across the agentive and non-agentive uses of EO psych verbs *bear different θ -roles*. In the non-agentive use, the indirect object must be more prominent than the subject argument interpreted as stimulus/subject matter and is not interpreted thematically as an agent, which we argue corresponds precisely to an experiencer based on the thematic hierarchy. On the other hand, in the agentive use, the indirect object must be thematically lower in prominence than the agent subject and given the asymmetry with the binding possibilities in the non-agentive use, must be lower in prominence than an experiencer (as in the non-agentive use) but higher than stimulus/subject matter, which we argue is therefore a theme. Assuming as well that the thematic hierarchy is reflected directly as structural hierarchy, if the indirect object argument outranks the subject argument in the thematic hierarchy, the subject argument must be base-generated lower than the indirect argument in the non-agentive use of EO psych predicates, which is precisely what is proposed in unaccusative approaches to EO psych verbs in general (Belletti & Rizzi 1988) and particularly for Chinese EO psych verbs (Cheung & Larson 2015), and this also directly explains the possibility of backward binding with the non-agentive variant.

Further evidence that the surface subject is lower on both the thematic hierarchy and structural hierarchy than the surface object in the non-agentive use of EO psych verbs comes

-
- (i) Wèile zìjǐ de lìyì shénmēilìliàng dōu zǔdǎng bù liǎo tā.
 for self DE profit what force all prevent NEG ASP he
 'No force can ever prevent him from working for his interest.'
 (Xu 1994: 118)

Xu argues that *he* in this sentence is an indirect agent as the English gloss shows, hence high on the thematic hierarchy. Nonetheless, in Xu's analysis, structural prominence still plays a crucial role in determining the binding relation: the subject DP, regardless of its thematic prominence, can always be the antecedent.

from a more interesting case involving multiple anaphors. We use the verb *gàosù* ‘tell’ again for illustration. In (43a), when the indirect object *Lǐsì* is the binder of the anaphor in the subject DP, *zìjǐ* in the embedded clause can no longer be bound by the subject DP, just as what is observed in (41b). Notably, the subject here must again be interpreted as a stimulus/source, meaning that it cannot be interpreted as Lisi’s rival agentively telling Lisi about Lisi’s own abilities, which is expected as the indirect object must be thematically more prominent than the subject as indicated by backward binding. By contrast, when the anaphor in the complement clause is bound by the subject, as shown in (43b), the coreference between *zìjǐ* embedded in the subject DP and *Lǐsì* becomes ungrammatical and the sentence is generally interpreted as Lisi’s rival agentively telling Lisi about his/her (the rival’s) own abilities. If we take the blocking of backward binding to be evidence that the surface subject is introduced high (external to vP), we can predict that *Lǐsì* in this case cannot bind the second *zìjǐ* as it is not thematically interpreted as an experiencer that is higher than the surface agent subject on the thematic hierarchy, and hence not as prominent as the agent to serve as the binder; expectedly, the non-culminating reading is available in this case.

- (43) a. [Zìjǐ_i de dùishǒu]_j gàosù Lǐsì_i [zìjǐ_{i/*j} hái bù gòugé],
 self DE rival tell Lisi self yet NEG qualified
 #dàn Lǐsì háishì yìwúsǔozhī.
 but Lisi still know.nothing
Intended: ‘(The winning of) his rival_j made Lisi_i realize that he_i is not qualified yet, but he still knows nothing.’
- b. [Zìjǐ_{*i} de dùishǒu]_j gàosù Lǐsì_i [zìjǐ_{*ij} hái bù gòugé],
 self DE rival tell Lisi self yet NEG qualified
 dàn Lǐsì háishì yìwúsǔozhī.
 but Lisi still know.nothing
Intended: ‘Lisi’s rival_j told Lisi_i that he_{*ij} is not qualified yet, but Lisi still knows nothing.’

We see here therefore that the facts about backward binding and binding by the more thematically prominent indirect object converge to show that agentive EO verbs and non-agentive EO psych verbs do not share the same structure. Furthermore, binding phenomenon in Chinese, which is sensitive to a thematic hierarchy, provides convincing evidence in support of our hypothesis that the objects across the agentive and non-agentive uses of most of the EO verbs do not bear the same θ -role, and the culminating reading in the non-agentive EO construction can be attributed to the thematic role of the object.

5. Conclusion

We examined in this paper EO psych verbs (e.g., *wēixié* ‘threaten’; *gǔlì* ‘encourage’; *ānwèi* ‘comfort’; *zhémó* ‘torture’; *xiūrǔ* ‘humiliate’) in Chinese with respect to the availability of non-culminating readings of the change in mental state that they describe in their event description. Like their much studied non-psych accomplishment counterparts, these predicates show a contrast between agentive and non-agentive subjects. With agentive subjects, the change in mental state on the part of the surface object is not entailed and defeasible, whereas non-agentive

subjects disallow non-culminating readings and the change in mental state must be entailed and non-cancellable. That is, EO psych verb predicates also show the agentive/non-agentive contrast, an effect that has been much discussed for accomplishment predicates cross-linguistically (Singh 1991, 1998; Koenig & Muansuwan 2000; Demirdache et al. 2017; Liu 2018; Beavers & Lee Juwon 2020; Martin et al. 2021; *a.m.o.*).

An influential explanation for the contrast, particularly in Chinese, comes from the work of Martin et al. (2020, 2021, 2023). They propose that there is a fundamental difference between agentive and non-agentive uses of accomplishment predicates in terms of how they are instantiated, with agentive uses providing an additional event, the agent's action, that can serve as input to a weak perfective marker, which optionally maximizes on this portion of the event excluding the change-of-state. Non-agentive uses, on the other hand, do not offer such an additional event and therefore, the entire change of state is the only valid eventive input to a weak perfective marker. Martin et al.'s (2020, 2021, 2023) account correctly predicts agentive uses of accomplishment predicates to allow non-culminating readings whereas non-agentive uses do not. Regarding psych verbs, they note that EO psych predicates in French and German also show the same agent non-agent contrast even though non-psych accomplishment predicates do not show it; for EO psych verbs, they propose the presence of a sub-lexical modal operator within a bi-eventive accomplishment event structure, and the contrast arises from the choice of modal base as determined by whether the subject is an agent or non-agent argument.

Nonetheless, we showed that none of the above accounts extend to Chinese EO psych predicates primarily because these EO psych predicates cannot reliably be shown to all behave like accomplishment predicates based on well-established diagnostics. This means that we cannot rely on any account that crucially requires causing and result sub-events. Furthermore, the contrast between agentive/non-agentive uses of the verb persist even in non-perfective contexts without perfective marking. Instead, we proposed following previous literature that EO psych verbs are structurally ambiguous based on the properties of the subject. Uses of these predicates with agentive subjects have a regular transitive structure where the agent argument is structurally introduced high while the object argument is introduced low and thematically interpreted as theme. Non-agentive uses, on the other hand, have the structure whereby both the surface subject and object are base-generated low, with the surface object structurally higher than the surface subject and the arguments are interpreted as experiencer and stimulus/subject matter respectively. Together with an understanding of the lexical semantic content of thematic roles, in particular the difference between experiencers and themes being whether or not the mental state of the referent is involved, we straightforwardly predict that non-agentive uses of EO psych predicates must describe a change in mental state that culminates whereas agentive uses can but need not. We show further that this analysis is backed up by independent empirical evidence in the form of backward binding asymmetries, where non-agentive uses of these predicates permit backward binding indicating that the surface subject is introduced below the object thereby permitting backward binding, whereas agentive uses disallow such backward binding. We argue further that binding asymmetries in double object constructions, whereby an indirect object can bind a reflexive in a complement clause only in the non-agentive and not the agentive use, demonstrate that the indirect object (the direct object in transitive EO psych predicates) must be different in prominence with respect to the subject argument across agentive and non-agentive uses, lending further support to the analysis that the source of non-culminating readings lies in

the thematic interpretation of the indirect (and also direct object in transitive cases) object in these ditransitive EO psych predicates. These findings from Chinese imply that from a theoretical point of view, we cannot attribute non-culminating readings of predicates describing changes-of-state (including mental states) across and even *within* languages to a single source and the independent properties of different verb classes need to be taken into account.

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